INCH-POUND
MIL-PRF-25962D
19 April 1996
SUPERSEDING
MIL-V-25962C
31 OCTOBER 1988

PERFORMANCE SPECIFICATION

VALVE, LIQUID OXYGEN DRAIN

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of valve for draining 70 and 300 pounds per square inch gage (psig) liquid oxygen converters.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are cited in sections 3 and 4 of this specification. This section does Pnot include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of these lists, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications. The following specifications form a part of this document to the extent specified herein. Unless otherwise specified, the applicable issues of these documents are those listed in the specific issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

BB-A-1034	Air, Compressed, For Breathing Purposes
BB-N-411	Nitrogen, Technical

(Unless otherwise indicated, copies of the above specifications are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, Pennsylvania 19111-5094.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/TICLA, 3001 Staff Drive, Suite 1AE1-101A, Tinker AFB, OK 73145-3036 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the applicable issues of the documents which have been adopted by the DoD are those listed in the specific issue of the DoDISS cited in the solicitation. Unless otherwise specified, the documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE AS4396	-	Fitting End - Bulkhead Flared Tube Connection, Design Standard
SAE AS8010	-	Aviator's Breathing Oxygen Purity Standard

(Application for copies should be addressed to: SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.20.1	-	Pipe Threads, General Purpose (Inch)

(Application for copies should be addressed to: ASME, United Engineering Center, 345 East 47th St., New York, NY 10017.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the materials meet or exceed the operational and maintenance requirements, and promote economically advantageous life cycle costs.

3.3 Materials. All materials shall be suitably treated to resist corrosion due to electrolytic decomposition, salt air, and any other atmospheric condition that may be encountered during operational use or storage (see 6.3). Materials that come into contact with liquid oxygen (LOX) shall not contain hydrocarbons, be non-impact sensitive and non-degradable, and shall have a brittle point below -297°F.

3.4.1 Dimensions. The valve shall conform to the general dimensions in Figure 1 for length, width, height, inlet and outlet locations, and handle size and location.

3.4.2 Threads. All pipe threads shall conform to ASME B1.20.1.

3.4.3 Markings. The word OUTLET and an arrow pointing in the direction of the outlet shall be stamped on the valve body. In addition, the words OPEN and CLOSED with an arrow pointing to these directions shall comply with Figure 1.

3.4.4 Weight. The weight of the valve shall not exceed 0.25 pound.

3.5 Environmental conditions. The valve shall operate in the following conditions:

(a) Temperature: -297°F to 260°F

(b) Vibration: 0.02 in. double amplitude from 500 to 3,000 cycles per minute.

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3.6 Performance.

3.6.1 General. The valve shall operate only when the handle is placed in the fully open or fully closed position, and shall not change operating conditions without handle actuation. In addition, the valve shall operate in any mounting position.

3.6.2 Flow rate. The valve shall allow a minimum flow rate of 150 liters per minute (L/min) when under 5 psig or less.

3.6.3 Leakage.

3.6.3.1 Outlet port. The outlet port shall not leak more than 0.02 L/min when pressurized to 70 psig, 300 psig, and 10 inches of water.

3.6.3.2 Valve. The valve body shall withstand an internal pressure of at least 500 psig.

3.6.4 Reliability. The mean cycles between failures (MCBF) of the valve shall not be less than 1,000 cycles (see 6.2).

3.6.5 Cleanliness. The valve shall be free from contaminants such as oils, grease, oxidizable metals, shredded paper, lint, or any other combustible fluid or material.

3.6.6 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on four valves. This inspection shall include all the tests indicated in 4.6.

4.3 Conformance inspection. Conformance inspection shall include the tests in 4.3.1 and 4.3.2.

4.3.1 Individual tests. Each valve shall be subjected to the tests in 4.6.1, 4.6.2, and 4.6.3.

4.3.2 Sampling tests. Two valves selected at random from each lot up to 500 shall be subjected to the tests in 4.6.4, 4.6.5, 4.6.6, 4.6.7, and 4.6.8.

4.4 Inspection conditions.

4.4.1 Temperature and pressure. Unless otherwise specified, tests shall be conducted at ambient conditions.

4.4.2 Test medium. Gas used in testing shall conform to SAE AS8010, Type I; air conforming to BB-A-1034, Grade A or C; nitrogen conforming to BB-N-411 Class I, Grade A; or liquid oxygen conforming to SAE AS8010, Type II.

4.4.3 Valve cleaning and drying. Each valve shall be thoroughly cleaned and dried after immersion in water.

4.5 Requirements cross-reference matrix. Table I provides a cross-reference matrix of the section 3 requirements tested or verified in the paragraphs below.

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TABLE I. Requirements cross-reference matrix

REQUIREMENT	VERIFICATION
3.1	4.2
3.3	4.6.1, 4.6.11
3.4.1	4.6.1
3.4.2	4.6.1
3.4.3	4.6.1
3.4.4	4.6.1
3.5	4.6.5, 4.6.6, 4.6.8
3.6.1	4.6.10
3.6.2	4.6.4
3.6.3.1	4.6.3
3.6.3.2	4.6.7
3.6.4	4.6.9
3.6.5	4.6.2

4.6 Tests.

4.6.1 Examination of product. Each valve shall be inspected to determine conformance to weight, thread size, markings, dimensions, materials, and handle operation.

4.6.2 Cleanliness. Valve surfaces that come in contact with oxygen shall be free of contaminants such as rust, scale, dirt, paint, oil, ink, hydrocarbons, and cleaning compounds. Cleanliness of the surfaces shall be satisfactorily demonstrated by industrially accepted test methods (see 6.4).

4.6.3 Outlet port leakage. With the valve closed, pressures of 70 psig, 300 psig, and 10 inches of water shall be applied to the inlet port. Leakage from the outlet port exceeding 0.02 L/min shall be cause for rejection.

4.6.4 Flow rate. With the valve open, not more than 5 psig shall be applied to the inlet port and the flow through the valve shall be 150 L/min.

4.6.5 High temperature operation. The valve shall be subjected to a temperature of 260°F for a minimum of 4 hours. While at this temperature, the valve shall be subjected to the test specified in 4.6.3.

4.6.6 Low temperature operation. With the valve open, liquid oxygen shall be applied to the inlet port to ensure a steady stream of liquid oxygen from the outlet port; then the valve shall be closed. This procedure shall be repeated three times. The valve shall then be subjected to a temperature of -65°F for a minimum of 4 hours. At the end of the 4-hour period, the valve shall be subjected to the test specified in 4.6.3. Any evidence of leakage is cause for rejection.

4.6.7 Valve leakage. With the valve open and the outlet port plugged, a pressure of 500 psig shall be applied to the inlet port. There shall be no valve body leakage. The valve shall then be subjected to the test specified in 4.6.3.

4.6.8 Vibration. The valve shall be subjected to vibration of a constant double amplitude of 0.02 inches. The vibration shall vary regularly from 500 to 3,000 to 500 cycles per minute with a constant rate of change in frequency. After 9-hours, while continuing vibration, the valve shall be subjected to the test specified in 4.6.3.

4.6.9 Reliability. Four samples shall be tested for reliability. The equipment shall be operated under cryogenic conditions simulating actual usage. Reliability of the equipment shall be expressed in terms of mean cycles between failures (MCBF) and shall be not less than 1,000 cycles. A cycle shall consist of operating the valve between the open and closed position with liquid oxygen at a pressure of 300 psig applied to the inlet port of the valve. Liquid must pass through the valve each cycle. Every 400 cycles, each valve shall be subjected to tests according to 4.6.3 and 4.6.7. The valves shall be cycled until failure. Any evidence of leakage or malfunction shall constitute a failure.

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4.6.10 Orientation. While the valve is in each three mutually perpendicular positions, it shall be subjected to the tests specified in 4.6.3. and 4.6.8.

4.6.11 Corrosion. A corrosion test or materials certification shall be used to demonstrate satisfactory compliance with requirement 3.3.

5. PACKAGING

5.1 General. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department or Defense Agency automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The valve covered by this specification is intended to be used to drain 70 and 300 psig liquid oxygen converters.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Item identification.

c. Issue of the DoDISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.2).

d. When first article is required (see 3.1).

e. Reliability (see 3.6.4).

f. Data required (see section 4).

g. Packaging requirements (see 5.1)

6.3 Materials. The corrosion resistant materials used satisfactorily in the past have been brass, bronze, phosphor bronze, 18-8 corrosion-resistant steel, and anodized aluminum alloys.

6.4 Cleaning. For past proven methods of cleaning see MIL-STD-1246 and MIL-STD-1359.

6.5 Subject term (key word) listing.
Liquid Oxygen
Valve

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Air Force - 99

Preparing activity:
Air Force - 71

Reviewer:
DLA -CS

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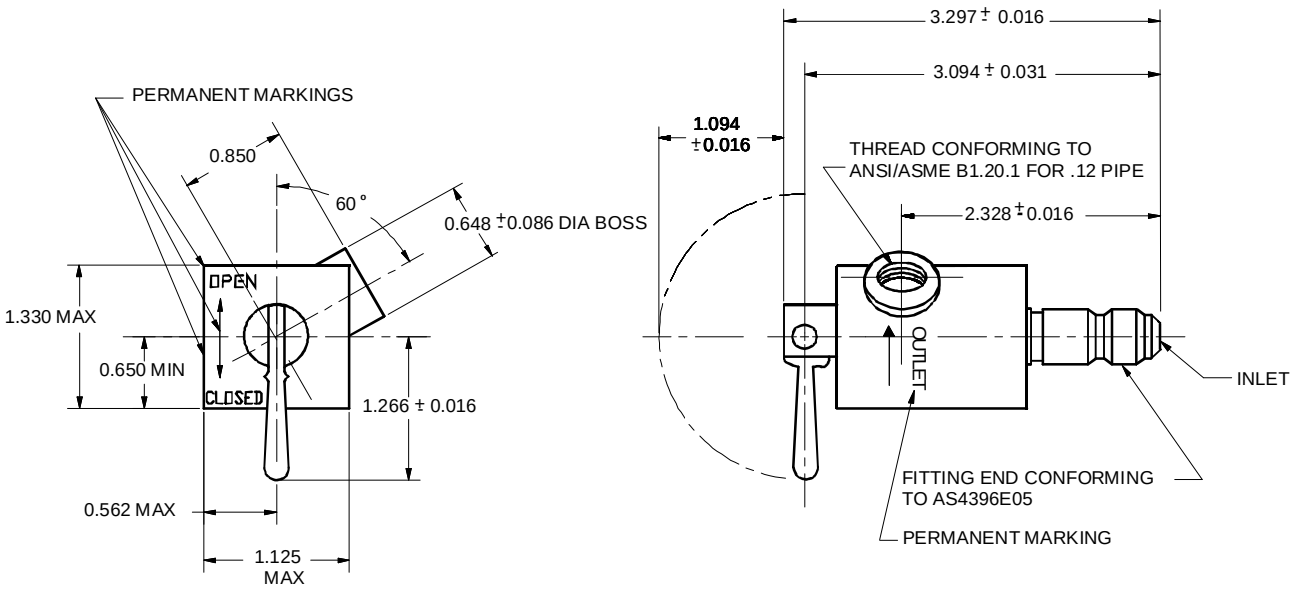

NOTE: UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES AND TOLERANCES ARE DECIMALS: ± 0.005, ANGLES ±1/2°

FIGURE 1. Valve envelope and port locations

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER**	**2. DOCUMENT DATE (YYMMDD)**
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3. DOCUMENT TITLE

VALVE, LIQUID OXYGEN DRAIN

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. Organization	
c. ADDRESS *(Include zip code)*	**d. TELEPHONE** *(Include Area Code)* (1) Commercial (2) DSN *(if applicable)*	**7. DATE SUBMITTED** **(YYMMDD)**

8. PREPARING ACTIVITY

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c. ADDRESS ***(Include Zip Code)*** ***OC-ALC/TICLA*** ***3001 Staff Drive Suite 1AE1-101A*** ***Tinker AFB, OK 73145-3036***	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** **Defense Quality and Standardization Office** **5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466** **Telephone (703) 756-2340/2343 DSN 289-2340/2343**

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